Partial Derivatives and the Gradient

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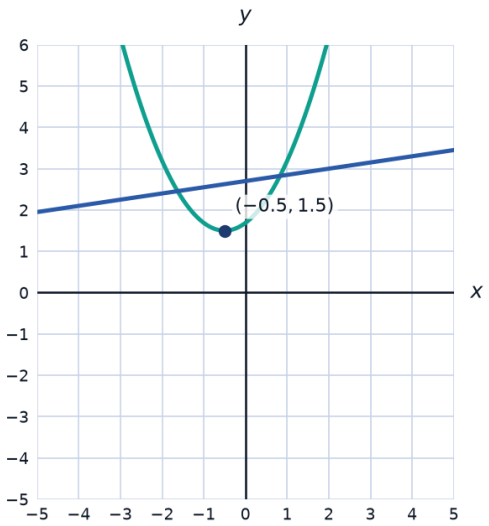
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: First and second partial derivatives

For $f(x, y) = e^{xy}\sin(x + y)$, find f_x using the product rule.

2 [Foundational] • Target: The gradient and directional derivatives

Find the gradient ∇f of $f(x, y) = x^2y + 3y^3$ at the point $(1, 2)$.



3 [Foundational] • Target: The gradient and directional derivatives

Find the directional derivative of $f(x, y) = x^2 + y^2$ at $(1, 1)$ in the direction of $\mathbf{v} = \langle 3, 4 \rangle$.

4 [Foundational] • Target: Tangent planes

Find the equation of the tangent plane to $z = f(x, y) = x^2 + y^2$ at the point $(1, 2, 5)$, using $z - z_0 = f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0)$.

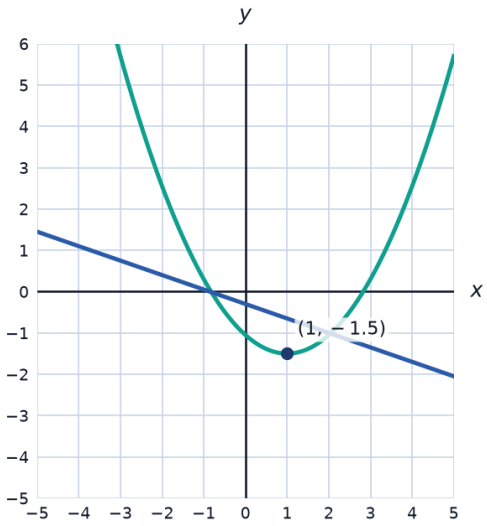
5 [Foundational] • Target: Tangent planes

Find the equation of the tangent plane to $z = xy$ at the point $(2, 3, 6)$.

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6 [Foundational] • Target: Critical points of multivariable functions

Find the critical point(s) of $f(x, y) = x^2 + y^2 - 4x + 2y + 5$ by setting $f_x = 0$ and $f_y = 0$.



7 [Foundational] • Target: Critical points of multivariable functions

Verify that the critical point found above is a minimum, using the second derivative test with $D = f_{xx}f_{yy} - (f_{xy})^2$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: The chain rule for multivariable functions

Let $z = x^2y$, where $x = t^2$ and $y = t^3$. Use the multivariable chain rule $\frac{dz}{dt} = \frac{\partial z}{\partial x}\frac{dx}{dt} + \frac{\partial z}{\partial y}\frac{dy}{dt}$ to find $\frac{dz}{dt}$ at $t = 1$.

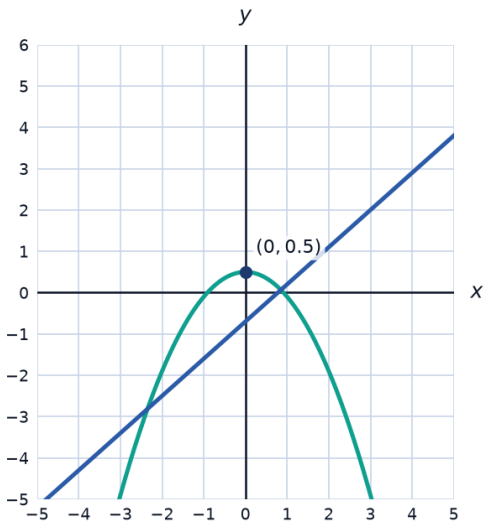
9 [Intermediate] • Target: The chain rule for multivariable functions

Let $w = x^2 + y^2 + z^2$, where $x = \cos t$, $y = \sin t$, $z = t$. Find $\frac{dw}{dt}$ using the chain rule.

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10 [Intermediate] • Target: Maximum rate of increase

Find the direction in which $f(x, y) = x^2 + xy$ increases most rapidly at the point $(1, 2)$, and state the maximum rate of increase.



11 [Intermediate] • Target: Maximum rate of increase

Find the direction of steepest descent for $f(x, y) = x^2 + y^2$ at the point $(3, 4)$.

12 [Intermediate] • Target: Implicit partial differentiation

For the surface $x^2 + y^2 + z^2 = 9$, find $\frac{\partial z}{\partial x}$ by implicit differentiation (treating y as constant).

13 [Intermediate] • Target: Classifying critical points with saddle points

Find and classify the critical point(s) of $f(x, y) = x^2 - y^2$, using the second derivative test.

14 [Intermediate] • Target: Classifying critical points with saddle points

Find and classify the critical point(s) of $f(x, y) = x^3 - 3xy + y^3$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

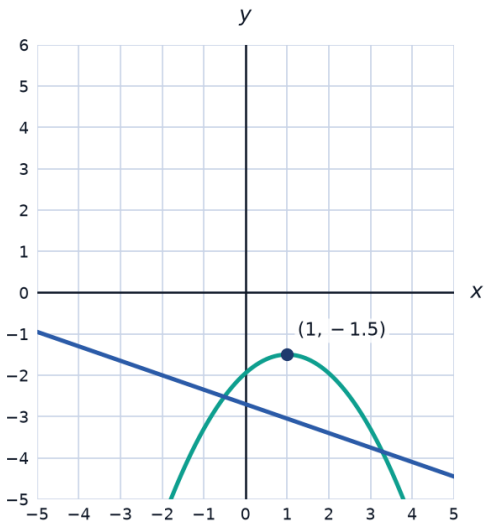
15 [Challenging] • Target: An applied optimization problem • Cross-Lesson: 8.02 Volumes of Revolution

A company's profit is modeled by $P(x, y) = 100x + 120y - x^2 - 2y^2 - xy$, where x and y are units of two products. Find the production levels that maximize profit by setting $P_x = 0$ and $P_y = 0$.

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16 [Challenging] • Target: Higher-order mixed partial derivatives • Cross-Lesson: 8.03 Arc Length and Improper Integrals

For $f(x, y) = x^3y^2 + \sin(xy)$, find f_{yx} and f_{xy}, and verify they are equal (as guaranteed by Clairaut's theorem for continuous mixed partials).



17 [Challenging] • Target: Linear approximation in two variables • Cross-Lesson: 9.01 Separable Differential Equations

Use the linear approximation $f(x, y) \approx f(a, b) + f_x(a, b)(x - a) + f_y(a, b)(y - b)$ to estimate $f(2.1, 2.9)$ for $f(x, y) = x^2y$, using the point $(2, 3)$.

18 [Challenging] • Target: Linear approximation in two variables • Cross-Lesson: 1.01 Precalculus Review for Calculus

The volume of a cylinder is $V(r, h) = \pi r^2 h$. A cylindrical can is manufactured with radius $r = 3$ cm and height $h = 10$ cm, but due to manufacturing tolerances, the radius might be off by as much as 0.05 cm and the height by as much as 0.1 cm. Use the total differential $dV = V_r\,dr + V_h\,dh$ to estimate the maximum possible error in the computed volume.

19 [Challenging] • Target: Linear approximation in two variables • Cross-Lesson: 10.01 Calculus with Parametric and Polar Curves

For $f(x, y) = x^2y - 3xy^2 + y^3$, find each of the following at the point $(1, -1)$:

a $f_x(1, -1)$

b $f_y(1, -1)$

c The gradient vector $\nabla f(1, -1)$

d The directional derivative in the direction of $\langle 1, 0 \rangle$

20 [Challenging] • Target: An applied optimization problem • Cross-Lesson: 11.01 Sequences and Series

Level 2 · Medium Find the maximum rate of change of $f(x, y) = 3x^2y - y^3$ at the point $(1, 2)$, and determine the unit vector in which it occurs.

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Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: First and second partial derivatives

Second-order Taylor approximation: $f(x,y) \approx f(a,b) + f_x(x-a) + f_y(y-b) + \frac{1}{2}[f_{xx}(x-a)^2 + 2f_{xy}(x-a)(y-b) + f_{yy}(y-b)^2]$. This involves:

a. only first partials

b. only second partials

c. third partials

d. both first and second partials

22 [Foundational] • Target: The gradient and directional derivatives

Chain rule with two paths: $z = f(x,y)$, $x = r\cos\theta$, $y = r\sin\theta$. $\partial z/\partial r =$

a. $f_x\cos\theta + f_y\sin\theta$

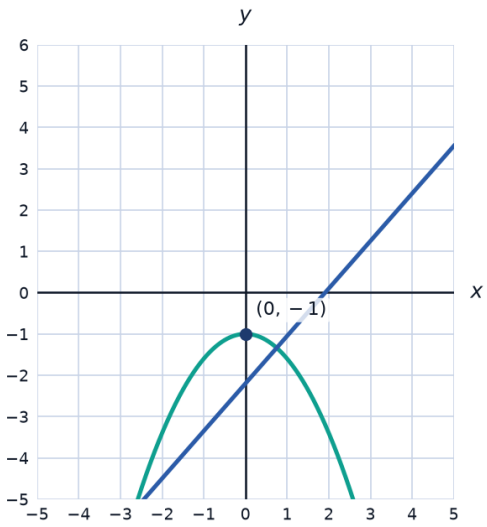
b. f_r

c. $f_x r + f_y r$

d. $f_x\sin\theta - f_y\cos\theta$

23 [Foundational] • Target: Tangent planes

Classify the critical points of $f(x,y) = x^2 - 4x + y^3 - 12y$:



a. $(2,2)$: saddle

b. $(2,2)$ min; $(2,-2)$ saddle

c. $(2,2)$: local min; $(2,-2)$: local max

d. $(2,2)$ saddle; $(2,-2)$ saddle

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24 [Foundational] • Target: Critical points of multivariable functions

Implicit function theorem for optimization: constrained critical points satisfy $abla_{\mathbf{x}} L = 0$ where $L = f - \lambda g$.

Name of λ:

a. Lagrange multiplier

b. eigenvalue

c. gradient

d. curvature

25 [Foundational] • Target: The chain rule for multivariable functions

Second mixed partial: for smooth f, $f_{xy} =$

a. f_{yx}

b. f_{yy}

c. f_{xx}

d. 0

26 [Foundational] • Target: Maximum rate of increase

Local extrema of f can occur only at critical points (interior) or on the:

a. axes

b. level curves

c. boundary

d. Hessian eigenvalues

27 [Foundational] • Target: Implicit partial differentiation

Find and classify all critical points of $f(x, y) = x^3 - 3x + y^3 - 3y$:

a. $(1, 1)$ and $(-1, -1)$ minima; $(1, -1)$ and $(-1, 1)$ saddles

b. $(\pm 1, \pm 1)$: all minima

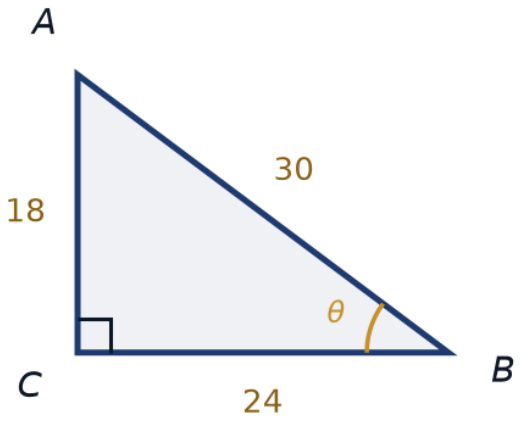
c. $(0, 0)$ only

d. no critical points

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28 [Foundational] • Target: Classifying critical points with saddle points

Tangent plane at $(2, 1)$ for $f = x^2y^3$: $f_x|_{(2,1)} = 4$, $f_y = 12$, $f(2, 1) = 4$. Approximation:



a. $4 + 4(x - 2) + 6(y - 1)$

b. $4 + 4(x - 2) + 12(y - 1)$

c. $8 + 4(x - 2)$

d. $4 + 8(x - 2) + 12(y - 1)$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: An applied optimization problem

The second derivative test fails ($D = 0$). This means the critical point is:

a. always a saddle

b. indeterminate without further analysis

c. always min

d. always max

30 [Intermediate] • Target: Higher-order mixed partial derivatives

Implicit differentiation: $F(x, y) = 0$. Then $dy/dx =$

a. F_y/F_x

b. $-F_x/F_y$

c. F_x/F_y

d. $-F_y/F_x$

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31 [Intermediate] • Target: Linear approximation in two variables

Second-order Taylor for $f(x,y)=e^{x+y}$ at $(0,0)$ up to second order:

a. $e(1+x+y)$

b. $1+x+y$

c. $1+(x+y)^2/2$

d. $1+x+y+\frac{1}{2}(x+y)^2$

32 [Intermediate] • Target: First and second partial derivatives

Linear approximation: $f(x,y)\approx f(a,b)+f_x(a,b)(x-a)+f_y(a,b)(y-b)$. This is accurate near:

a. (a,b) if f is differentiable

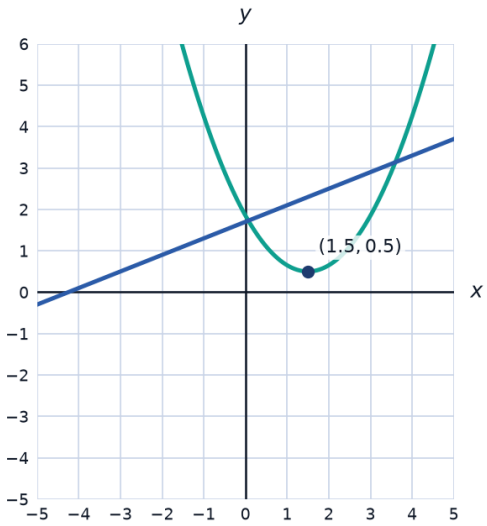
b. far from (a,b)

c. only when f is quadratic

d. when $ablaf=0$

33 [Intermediate] • Target: The gradient and directional derivatives

If $D<0$, the critical point is a:



a. local min

b. local max

c. saddle point

d. degenerate

34 [Intermediate] • Target: Tangent planes

Find the maximum of $f(x,y)=xy$ on the ellipse $\frac{x^2}{4}+\frac{y^2}{25}=1$.

a. 5

b. 10

c. $\frac{5}{2}$

d. $\frac{7}{2}$

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35 [Intermediate] • Target: Critical points of multivariable functions

Implicit function theorem: at (a, b), $\partial y/\partial x =$

a. $D_x F$

b. $-(D_y F)^{-1} D_x F$

c. 0

d. I

36 [Intermediate] • Target: The chain rule for multivariable functions

$f(x, y) = xy$: at origin, the second derivative test gives $D = f_{xx} f_{yy} - f_{xy}^2 = 0 - 1 = -1 < 0 \Rightarrow$

a. saddle point

b. local max

c. local min

d. inconclusive

37 [Intermediate] • Target: Maximum rate of increase

Euler's theorem for homogeneous functions: if $f(tx, ty) = t^n f(x, y)$, then $x f_x + y f_y =$

a. f

b. nf

c. $n(n-1)f$

d. 0

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Implicit partial differentiation • Cross-Lesson: 1.01 Precalculus Review for Calculus

For $f(x, y) = 4x^2 + 3y^2 - 8x - 6y + 4$: critical point and value:

a. $(1, 2)$ with $f = -1$

b. $(1, 1)$ with $f = 0$

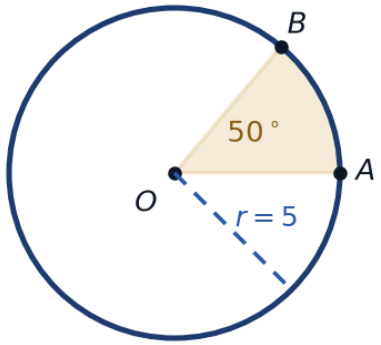
c. $(2, 1)$ with $f = 0$

d. $(1, 1)$ with $f = -3$

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39 [Challenging] • Target: Classifying critical points with saddle points • Cross-Lesson: 10.01 Calculus with Parametric and Polar Curves

Find the maximum of $f(x, y) = 5x + 12y$ on the circle $x^2 + y^2 = 25$.



a. 85

b. 325

c. 65

d. 169

40 [Challenging] • Target: An applied optimization problem • Cross-Lesson: 11.01 Sequences and Series

Find the tangent plane to the surface $x^2 + y^2 - z = 4$ at $(1, 2, 1)$:

a. $2x + 4y + z = 9$

b. $x + 2y - z = 4$

c. $2x + 4y - z = 9$

d. $x + 2y = 5$

41 [Challenging] • Target: Higher-order mixed partial derivatives • Cross-Lesson: 12.01 Convergence Tests

Find the tangent plane to $z = \ln(x^2 + y^2)$ at the point where $(x,\ y) = (1,\ 0)$.

a. $z = 2x - 2$

b. $z = 2y$

c. $z = 2 - 2x$

d. $z = 2x$

42 [Challenging] • Target: Linear approximation in two variables • Cross-Lesson: 12.02 Power Series and Taylor Series

Classify the critical point $\left(-\frac{24}{47},\ \frac{3}{47}\right)$ of $f(x, y) = 3x^2 + xy + 3x + 4y^2$.

a. local maximum

b. saddle point

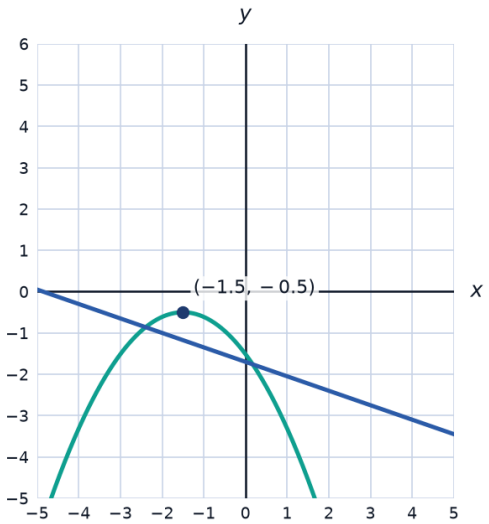
c. the test is inconclusive

d. local minimum

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43 [Challenging] • Target: First and second partial derivatives • Cross-Lesson: 13.01 Vectors and Vector-Valued Functions

Directional derivative $D_{\hat{u}}f = ablaf \cdot \hat{u}$. For $f = xy$, $\hat{u} = (1/\sqrt{2}, 1/\sqrt{2})$, at $(1, 1)$:



a. 2

b. 1

c. 0

d. $\sqrt{2}$

44 [Challenging] • Target: The gradient and directional derivatives • Cross-Lesson: 15.01 Double Integrals

Find the maximum of $x + y + z$ on the sphere $x^2 + y^2 + z^2 = 3$.

a. 3

b. 9

c. $\sqrt{3}$

d. $3\sqrt{3}$

45 [Challenging] • Target: Tangent planes • Cross-Lesson: 16.01 Vector Fields and Line Integrals

Maximum of xy on ellipse $x^2/4 + y^2/9 = 1$ by Lagrange:

a. 3/2

b. 3

c. 9/4

d. 6

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